

Question

A and B are metallic elements, with B being a transition metal, and X is a non-metallic element; A, B, and X form a salt with the chemical formula A_2BX_4 , which does not contain water of crystallization. A_2BX_4 decomposes upon heating in a hydrogen atmosphere, undergoing an endothermic reaction at 365°C ; further heating to 515°C results in a weight loss of 12.8%; continued heating to 930°C leads to an additional weight loss of 51.9%, leaving a residual elemental solid. Heating A_2BX_4 to 600°C in a hydrogen atmosphere results in a total weight loss of 25.7%, with the remaining solid being the alloy A_2B . Calculate the sum of the atomic numbers of the three elements A, B, and X.

A. 133

B. 160

C. 134

D. 114

E. 142

F. 82

G. 73

H. 105

Answer

Correct Answer: **E**

Detailed Explanation

From the second equation of weight loss in the problem, the following relationship can be obtained:

$$(1-25.7\%) * [2M(A) + M(B) + 4M(X)] = 2M(A) + M(B)$$

CHECKPOINT

1 PTS

List the weight loss equation: $(1-25.7\%) * [2M(A) + M(B) + 4M(X)] = 2M(A) + M(B)$

From the first equation of weight loss, we get: the remaining elemental metal should be one of A or B. Therefore, list the weight loss equation:

$$(1-12.8\%-51.9\%) * [2M(A) + M(B) + 4M(X)] = M(B) \text{ or}$$

$$(1-12.8\%-51.9\%) * [2M(A) + M(B) + 4M(X)] = 2M(A)$$

CHECKPOINT

1 PTS

List the weight loss equation: $(1-12.8\%-51.9\%) [2M(A) + M(B) + 4M(X)] = M(B) \text{ or } (1-12.8\%-51.9\%) [2M(A) + M(B) + 4M(X)] = 2M(A)$

Given that X is a non-metallic element, substitute the molecular weights of several common non-metallic elements such as Cl, O, and S to obtain the following results:

When $M(X) = 16$:

$M(A) = 48.56$, $M(B) = 87.91$ or

$M(A) = 43.95$, $M(B) = 97.12$

When $M(X) = 32$:

$M(A) = 97.12$, $M(B) = 175.81$ or

$M(A) = 87.91$, $M(B) = 194.2$

When $M(X) = 35.5$:

$M(A) = 107.74$, $M(B) = 195.04$ or

$M(A) = 97.52$, $M(B) = 215.49$

It is found that when X is Cl and the residual metal is B, there is a reasonable solution.

CHECKPOINT

2 PTS

X=Cl

It is calculated from $Cl = 35.5$ g/mol that the molecular weight of A is 108 and the molecular weight of B is 195. Then A is Ag and B is Pt. Therefore, the sum of the atomic numbers of the three elements is: $47+78+17=142$, so choose E.

CHECKPOINT

1 PTS

B=Pt

CHECKPOINT

1 PTS

 $A=Ag$

CHECKPOINT

1 PTS

 $47+78+17=142$